TRIAL PSPM SM015

Name:	
Class:	
Lecturer:	

LIST OF MATHEMATICAL FORMULAE

Trigonometry

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$$

$$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$$

$$\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$$

$$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$$

$\sin 2A = 2\sin A \cos A$

$$\cos 2A = \cos^2 A - \sin^2 A$$
$$= 2\cos^2 A - 1$$
$$= 1 - 2\sin^2 A$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

Differentiation

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule

$$\frac{d}{dx}\big(u(x) + v(x)\big) = \frac{du}{dx} + \frac{dv}{dx}$$

Product Rule

$$\frac{d}{dx}\big(u(x) \cdot v(x)\big) = v\frac{du}{dx} + u\frac{dv}{dx}$$

Quotient Rule

$$\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Chain Rule

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \times \frac{dt}{dx}$$

Shape	**Volume**	**Surface Area**
Sphere	$V = \frac{4}{3}\pi r^3$	$S = 4\pi r^2$
Right circular cone	$V = \frac{1}{3}\pi r^2 h$	$S = \pi r^2 + \pi r h$
Right circular cylinder	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$

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SECTION A [25 marks]

This section consists of **3** questions. Answer **all** the questions.

1. Evaluate the following limits.

(a) $\lim\limits_{p \to 2} \dfrac{p^2 - 4}{8 - p^3}$.

[5 marks]

(b) $\lim\limits_{x \to 4} \dfrac{x - 4}{\sqrt{2x-1} - \sqrt{x+3}}$.

[4 marks]

2. (a) Find the derivative of $f(x) = 2\sqrt{x}$ using the first principle.

[5 marks]

(b) Find $\dfrac{dy}{dx}$ for $y = \ln\left(x^2\sqrt{1-3x}\right)$.

[4 marks]

3. Sand is leaking through a small hole at the bottom of a conical funnel at the rate 6 cm^3 / s. If the radius of the funnel is 4 cm and the height of the funnel is 8 cm, find the rate at which the sand depth decreases when it is 3 cm inside the cone.

[7 marks]

SECTION B [75 marks]

This section consists of **7** questions. Answer **all** the questions.

1. Without using a calculator, express $\dfrac{3+2i}{(1-4i)i^3}$ in the form of $a + bi$.

Hence find the exact value of $\left|\dfrac{3+2i}{(1-4i)i^3}\right|$.

[5 marks]

2. (a) If $\log_a\left(\dfrac{x}{a^2}\right) = 3\log_a 2 - \log_a (x - 2a),$ express x in term of a.

[5 marks]

(b) Solve $\dfrac{24-8x}{x^2-9} \geq -2$.

[6 marks]

3. Functions g and $f \circ g$ are defined by $g(x) = e^{x+2}$ and $(f \circ g)(x) = \sqrt{4x-1}$ for all $x \geq 0$.

   (a) Find the function f and determine its domain.

   [5 marks]

   (b) Determine the value of $(g \circ f)(e^3)$.

   [2 marks]

4. Given $f(x) = \dfrac{3}{1-x}$, $x > 1$.

   (a) Find all x such that $f(x) > 1$.

   [2 marks]

   (b) Show algebraically that $f(x)$ is a one-to-one function.

   [3 marks]

   (c) Determine the inverse function of $f(x)$.

   [3 marks]

   (d) State the domain and range for $f^{-1}(x)$.

   [2 marks]

   (e) Sketch the graph $f(x)$ and $f^{-1}(x)$ on the same axes.

   [3 marks]

5. Consider

$$w(x) = \begin{cases} x^2 + 2x - 5m + 9 & , x < 0 \\ e^x + 3m & , 0 \leq x < 2 \\ \dfrac{3x-2}{x+4} & , x > 2 \end{cases}$$

   (a) Find $\lim_{x \to 0^+} w(x)$ and $\lim_{x \to 0^-} w(x)$. Hence, determine the value of m such that the function $w(x)$ is continuous at $x = 0$.

   [5 marks]

   (b) Determine whether $w(x)$ is continuous at $x = 2$. State your reason.

   [2 marks]

   (c) Determine the horizontal asymptote for $x > 2$.

   [3 marks]

   (d) Sketch the graph for $w(x)$. Hence, find the range of the function.

   [4 marks]

6. (a) Given $y^2 \sin x = \cos y^3$, find $\dfrac{dx}{dy}$.

[6 marks]

(b) A curve is given parametrically by equation $x = e^{-2t}$ and $y = te^{-t}$.

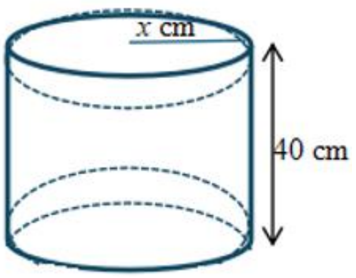
(i) Find the coordinates of the stationary point.

[7 marks]

(ii) Determine the point of stationary point is maximum or minimum.

[3 marks]

7. Two hemispheres are inscribed at the end of a cylinder of radius x cm and height 40 cm to form a solid as shown in the diagram.



(a) Show that the volume of the solid is $V = 40\pi x^2 - \dfrac{4}{3}\pi x^3$.

[3 marks]

(b) Find the radius in such that a way the volume of the solid is maximum. Hence, calculate the exact maximum volume.

[6 marks]

END OF QUESTION PAPER